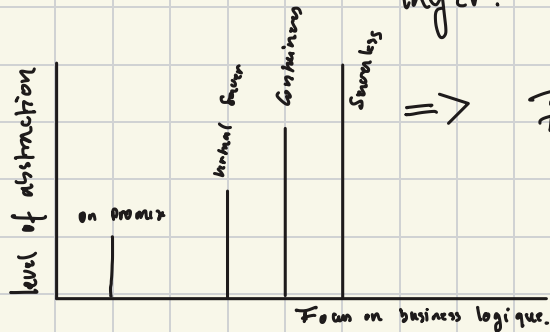




Lambda

⇒ Cloud Computing : allow to abstract the infrastructure layer.



⇒ Focus on business logic. : Small pieces loosely joined

⇒ Lambda is a Serverless Compute Service that you can use to run code without provisioning or managing servers.

- Invokes your code in response to events
- Scales automatically
- provides built-in code monitoring and logging with amazon Cloudwatch.

⇒ Pay only for the time the code is executed.



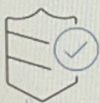
Offers the ability to bring your own code



Integrates with and extends other AWS services



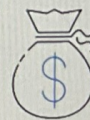
Offers flexible resource and concurrency models



Offers a flexible permissions model



Provides built-in availability and fault tolerance



Reduces the need to pay for idle resources

+ Lambda function



↳

Access permissions → matching events → App code & dependencies → runtime and deployment package → Config.

+ Concurrency: The number of function invocations that are running at one time.
 $\swarrow$ Reserved Concurrency
 $\searrow$ regional Concurrency

Invoking a lambda function.

① **Push Event** ⇒ An AWS service directly invokes the lambda function.

② **Pull (polling event)** ⇒ Lambda polls for events and invokes the lambda function with batches of records. ⇒ an event put the information into a stream or queue. Lambda pulls the streams or queue and invokes the function when it detects an event.

Push → 2 sous-types

	Synchrone	Asynchrone
Comportement	L'appelant attend la réponse	Lambda queue l'événement, l'appelant repart immédiatement
API Invoke type	RequestResponse	Event
HTTP status retourné	200	202 (accepted)
Retry	Aucun built-in — c'est ton code qui gère	0-2 retries automatiques
Exemples sources	API Gateway, ELB, Cognito, Alexa	S3, SNS, SES, CloudWatch Events

⚠ Piège exam : tu n'as aucun contrôle sur le type d'invocation quand c'est un service AWS qui trigger Lambda. C'est prédéfini par le service source.

Pull (polling) → 2 sous-types

Stream-based (DynamoDB Streams, Kinesis) :

- Lambda utilise un Event Source Mapping pour poller les shards
- En cas d'échec → Lambda bloque la lecture de ce shard jusqu'à ce que le batch soit traité ou expire (24h)
- Risque : un record en erreur peut bloquer tout le stream → configurer une OnFailure destination pour bypass

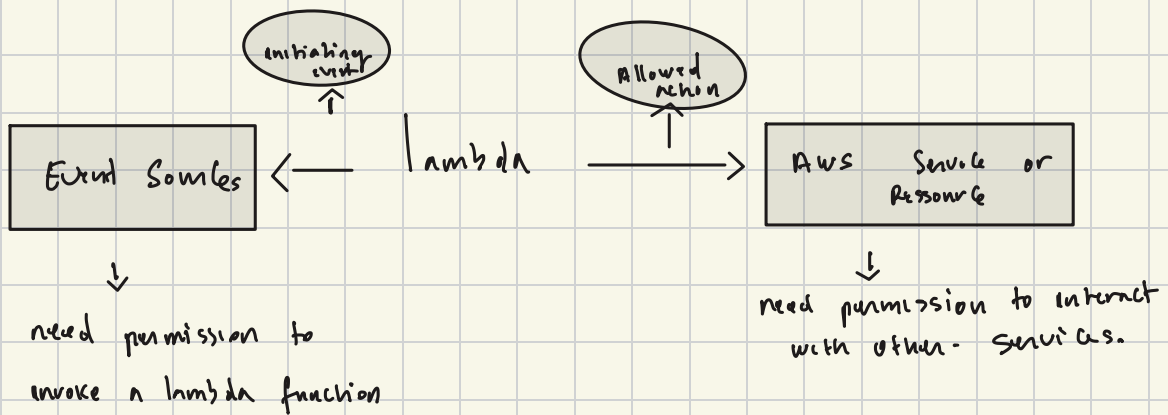
Queue-based (SQS) :

- En cas d'échec → le message redevient visible dans la queue, retenté jusqu'au retry limit
- Failed messages → Dead Letter Queue (DLQ)

Note coût : le polling lui-même ne coûte rien. Tu paies seulement si des actions résultent du poll.

⇒ Practical Example for the Exam.

Permissions



① An IAM Resource policy gives permissions to invokes the function.
 e.g: the resource policy will be in the bucket and not the lambda function.

② Lambda execution role specifies what the lambda function is permitted to do.

→ IAM policy and Trust policy

↓
 Définit les permissions
 vers d'autres services.
 → C'est qui l'a le droit de faire.

↓
 Définit quelle entité a le droit
 d'assumer le rôle IAM. Sans ça même si
 le rôle existe, personne ne peut l'utiliser.
 ⇒ qui peut entre ?

Deployment options for lambda

• Zip

→ choose a runtime when we create the function and upload them to lambda

• Container Image.

→ choose a runtime and linux distribution when creating the image. Package the code and upload the image. (ECR)

↳ Uploading Zip: Check size 
 Package deployment

↳ Lambda Console Editor
 upload the deployment package directly from IDE.
 Compress Zip and upload to the S3 bucket.

⇒ Lambda Support Versioning

‡ Last: Default version (1st)
 Alias: pointeur ⇒ dev, test, Prod.

- Une fois la fonction publiée $\Rightarrow$ Can't be updated $\Rightarrow$ Immutable

- Layer : Code partagé entre les fonctions : Exemple : SO M de dépendances.

$\Rightarrow$ Relier les services à l'aws et non directement à la version de lambda. $\rightarrow$ lors d'un new deploy. il faut modifier le Config de chaque services ! $\Rightarrow$ avec

Monitoring

+ Cloud watch : Provides built-in logging and metrics to monitor your lambda functions.

+ X-Ray : We can trace requests across your application to find trouble spots quickly.